Simulator calculation method

Independent student analysis | Model 7.2.1 | Report case: Hub & Book, pilot first

What the simulator calculates

We compare each investment with continuation of the existing service, using the same starting customers, external demand and time horizon. Customers move monthly between occasional, frequent, Pass and lapsed states, with baseline replacement inflow. The report case uses a derived Oslo scale estimate of about 125,000 active customers, a 36-month horizon, a 10% annual discount rate and no terminal value.

From customer behaviour to activity

Booking attempts = (eligible frequent customers x their adoption + eligible Pass customers x their adoption) x bookings per adopter x rollout/ramp exposure x monthly demand. Adoption changes with price. Fulfilment, no-shows and any Pass allowance determine delivered rides and fee revenue. Group trips follow occasions x initiation x vehicle availability x completion; additionality separates new trips from existing trips. Guest activation and incentives use their entered assumptions.

From activity to incremental contribution

Revenue is net of VAT. The calculation includes ride contribution, booking fees and additional customer activity, less placement, support, compensation, payment, incentive and implementation costs. Lost ordinary rides are charged to the proposal; shared programme work is counted once. All priced resource costs enter economic value. Cash expenditure is tracked separately: reassigned staff time can remain an economic cost without being new payroll. For example, a NOK 35 fee yields NOK 28 before costs after 25% VAT. Placement costs 8/60 x NOK 400 + NOK 20 = NOK 73.33 per job; 20% placement therefore costs NOK 14.67 per attempt before overtime.

Value, funding and payback

For each month, subtract baseline economic contribution from proposal contribution, including implementation costs when incurred. Discount those differences and sum them over the horizon. Peak funding is the largest cumulative incremental cash deficit within that horizon. Discounted payback is the first month cumulative discounted economic contribution reaches zero; inspect later months for renewed losses.

Value = sum of [proposal contribution(t) - baseline contribution(t)] / (1 + r)^(t / 12), for t = 0 to H.

Implementation and decisions

The report path launches booking in month 4 and tests expansion after the month-9 gate. Its headline includes expansion from month 10; stopping at the gate is calculated separately. Direct rollout uses its own cost timing on the same baseline. At the gate, compare future contribution with remaining avoidable costs, excluding sunk expenditure, and check delivery, operating rights and funding. The proposed report screen is value / peak funding of at least 1.0 and discounted payback within 24 months; absolute value remains the simulator's ranking basis.

Worked inputs: NOK 35 fee, eight bookings per adopter, 40% eligible reach, 95% fulfilment and 20% placement. At NOK 35, effective assumed adoption is 15% of eligible frequent customers and 30% of eligible Pass customers.

Worked report case	Result
Incremental discounted value	NOK 0.954m
Maximum cumulative cash need	NOK 0.623m
Value per NOK of peak cash / discounted payback	1.53 / month 21

How to interpret the result

Adoption, frequency, reach, placement and price response are argued assumptions to be tested. Survey needs do not estimate paid demand. The report case credits no unmeasured retention, repeat-use, upgrade or placement-spillover uplift. Sensitivities show where conclusions change; uncertainty shares describe the chosen assumed distributions. These are conditional investment scenarios.